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RESEARCH-ARTICLE

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Internet of things (IoT) development tends to reduce the reliance on centralized servers. The zero-trust distributed system combined with blockchain technology has become a hot topic in IoT research. However, distribution data storage services and different blockchain protocols make network interoperability and cross-platform more complex. Relay chain is a promising cross-chain technology that solves the complexity and compatibility issues associated with blockchain cross-chain transactions by utilizing relay blockchains as cross-chain connectors. Yet relay chain cross-chain transactions need to collect asset information and implement asset transactions via two-way peg. Due to the release of user transaction information, there is the issue of privacy leakage. In this article, we propose a cross-chain privacy protection protocol based on the Groth16 zero-knowledge proof algorithm and coin-mixing technology, which changes the authentication mechanism and uses a combination of generating functions to map virtual external addresses in transactions. It allows fast cross-chain anonymous transactions while hiding the genuine user's address. The experiment shows that, in a zero-trust IoT context, our scheme can effectively protect user privacy information, accomplish controlled transaction traceability operations, and guarantee cross-chain transaction security.

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1 INTRODUCTION

1.1 Background

The **Internet of Things (IoT)** is currently reaching new levels with approximately 5.8 billion IoT end devices as of 2020. The global IoT device value will reach approximately $120 million by 2022. And an expected of IoT devices will increase to 75 billion by 2025. IoT device management and data storage will become more complicated. Zero-trust network models and blockchain technologies are becoming popular in IoT research directions [2, 25].

The zero-trust network model is significant in heterogeneous ecosystems of smart devices with complete distrust and decentralization. It provides a dynamic access control policy independent of network location for network security [24]. A zero-trust network for IoT based on blockchain technology will not rely on network location and traffic data for security. Instead, all communications are subject to fine-grained access control and traffic analysis to prevent privacy leakage, utilizing blockchain network autonomy and cryptographic authentication to more effectively protect data security and user privacy [18, 22, 28].

IoT devices currently in use come in a wide variety. Many IoT terminals, such as smart homes, industrial sensors, smart cars, and so on, produce a lot of data. Blockchain and IoT integration create a secure, open, and controlled data processing platform [1, 31, 32, 37]. Numerous blockchain frameworks for different applications have emerged as a result of the rapid developments of blockchain technology. These blockchains can be divided into three main groups based on how open they are (public blockchains, hybrid blockchains, and private blockchains) [4], which use different standards, consensus algorithms, and asset values. Information cannot be transferred between blockchains, because the blockchain networks are segregated from others and form separate parallel networks. Blockchain cross-chain technology is proposed to resolve the issue of information isolation.

Cross-chain is primarily a technology for establishing bridges between different blockchains to achieve value exchange or value transfer [43]. It utilizes standard cross-chain protocols to ensure atomicity, consistency, and interoperability. The Hash Time Lock Mechanism [40], Notary Scheme [39], Side Chain or Relay, and Distributed Private Key Control [14] are examples of cross-chain mechanisms that have been validated.

The purpose of the hash time lock mechanism is primarily to establish a transaction channel, in which the two parties will use the respective assets locked in the intermediary address using hash lock and time lock, with the recipient unlocking the assets after the transaction is successful. The notary mechanism verifies the consistency and legitimacy of transactions and relies on the third party to complete cross-chain transactions by introducing a third-party trading institution. Distributed private key control is a key distribution mechanism that maps the original chain assets to a new blockchain and allows for the exchange of digital assets on the new chain. The Side

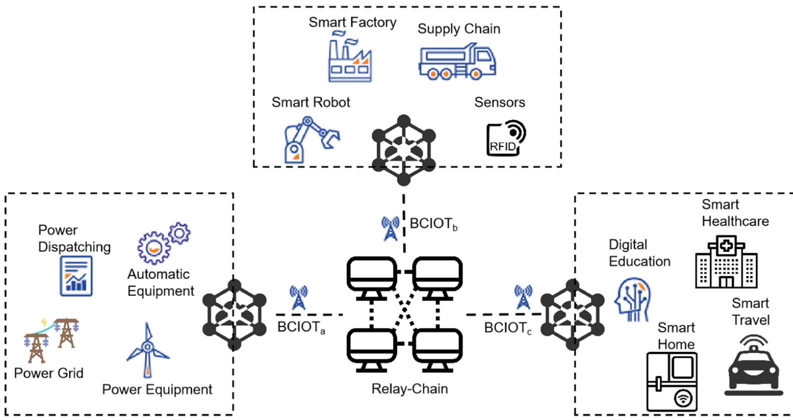

Fig. 1. The architecture of cross-chain based on zero trust BCIOT.

Chain or Relay model is mostly used to complete transactions by collecting state information from different blockchains and locking assets using two-way pegs [20].

Current cross-chain frameworks have gradually developed many commercial standards, such as BTC-relay, Cosmos, and Polkadot [43], which have mature applications in public blockchains BTC and ETH. There have also been many cross-chain frameworks with new technological advancements in recent years of research. Xu et al. [40] proposed a hash-locked cross-chain contract framework HTLCs that utilizes game theory and collateral model improvements to improve the success rate of transactions. Ye et al. [30] proposed an inter-chain messaging protocol, IBTP, and a generic high-speed cross-chain framework, BitXhub, based on a side-chain relay chain architecture, which has a more generic cross-chain protocol and a well-established cross-chain standard. Frauenthaler et al. [15] improved relay chains in ETH, simplified payment validation, and improved cross-chain protocols, making cross-chaining easier and more efficient. Madine et al. [26] proposed appXchain, a blockchain interoperability solution based on decentralized applications, which is fully decentralized and low-cost to deploy.

In these cross-chain frameworks, more attention pay to the speed and versatility of the cross-chain, not security. The relay chain collects user transaction information to ensure that the amount of assets traded is accurate and that transactions are genuine and valid. As a result, permission to share sensitive user information is impossible to control. Especially in the cross-chain frameworks based on relay chains and notaries, autonomous systems with zero trust have privacy leakage problems when interconnecting.

1.2 Problems

IoT plays a significant role in the industry, energy, and social services. Particularly in a zero-trust environment dominated by the **Blockchain Internet of Things (BCIOT)**, where different organizations or regional institutions form isolated BCIOT from one another. We can use cross-chain technology to solve the problem of information sharing and interoperability [19, 29, 33, 34].

The overall architecture of the IoT cross-chain framework with relay chain is depicted in Figure 1. Multiple ends of the IoT blockchain network are connected by an intermediary chain, and a cross-chain system can be swiftly constructed by deploying smart contracts with strong generality. The main process of its asset transaction is as follow: User A initiates a cross-chain transaction and locks it on BCIOT$_a$. The locking operation is to send this transaction asset to a common address. The relay chain will acquire the transaction information of BCIOT_a and obtain the SPV transaction proof. After the consensus verifies the correctness, an unlocked transaction is created in BCIOT_b. The relay chain is anchored in BCIOT_a. The asset of BCIOT_b will be updated, and User B will receive the unlocked transaction [9, 23].

The transaction process of BCIOT_a user A and BCIOT_b user B will be recorded by the relay chain and broadcast to other nodes. The relay chain to verify the block header information of the two chains. If there is a malicious node to collect user information and form a user portrait, then attackers will maliciously target users or engage in unfair competition with them.

The standard approach to cross-chain privacy protection typically employs a public address or UTXO to conceal the transaction address. Both parties in cross-chain transactions require an intermediary for credit endorsement and asset proof. When directly using coin-mixing transactions or zero-knowledge proofs [21], it is impossible to notarize the transaction and trace the transaction with anonymous transaction records, making management and control difficult. Transactions between businesses ought to be stable and controllable and not reveal confidential business information to the right holders.

1.3 Contributions

In this article, we focus on the security of personal asset information and transactions when using relay chains as a cross-chain bridge in a hybrid chain-based IoT blockchain and realize cross-chain transactions without revealing specific assets and the behavior of both parties to the transaction. We utilize multiple public address pools, and the internal address of the transaction will be assigned to a common address. The virtual address generator will generate a discrete set of points in the sine function to select a point and use the snowflake algorithm to compose a 64-bit address, which we call the external address. It is used for one-way lookup and transaction traceability. The final transaction will use the groth16 algorithm [16] to create a definite credential and finish it.

A privacy-preserving framework for cross-chain transactions based on zero-knowledge proofs using virtual addresses is defined in this article. It relies on smart contracts of the blockchain and public address pools to complete the transactions. According to each other's contract rules, we can infer that if both parties cannot agree, then the transaction cannot be completed and must wait for the next initiated transaction. Our solution can guarantee the anonymity and correctness of the transaction. It is safer and more efficient compared to other privacy protection schemes.

The contribution of the cross-chain privacy protection scheme in this article is summarized as follows:

- We propose a blockchain cross-chain communication protocol running in a zero-trust IoT environment for the protection scheme of privacy leakage during the cross-chain process, which is based on relay chains and ensures the security of inter-chain transactions by hiding transaction addresses to shield key information.
- We change the groth16 algorithm verification logic so that it can generate an additional determining public parameter to verify the prover's polynomial. This non-interactive verification process will expedite the verification of cross-chain transaction accounts and amounts, ensuring transaction accuracy.
- We provide an address generator to map each user's transaction to a different external address, which can be computed uni-directional using a generation curve unique to each chain so that users from other chains are unaware of the transaction mapping relationship.
- Through experiments and security analysis, we demonstrate that our cross-chain solution can enable two parties to collaborate for transaction traceability while ensuring privacy and security and that it can be implemented quickly using smart contracts and relay routing.

To this end, the rest of this article is organized as follows. In Section 2, we introduce the results of some recent papers that do privacy protection in cross-chain. In Section 3, we elaborate on our scheme for cross-chain privacy leakage and propose a cross-chain protocol. The security analysis for the cross-chain protocol put out in this article is presented in Section 4. We design experiments in Section 5 to evaluate the effectiveness of the cross-chain interaction procedure. Finally, we summarize this article and the outlook for future work in Section 6.

2 RELATED WORK

In recent years, there have been many security optimization schemes in the cross-chain field, each with a different focus. In this section, we will show some work on privacy protection in cross-chain technology in a systematic way.

Traditional atomic exchanges ensure transaction correctness but do not account for the anonymity of transaction records and the transaction process, which can analyze the smart contract to retrieve transaction information and lock the addresses of both parties to the transaction. Reference [5] presents a new atomic exchange strategy that newly adds an anonymous transaction to the blockchain using the UTXO unconsumed model. Zero-knowledge proofs are used to offer self-authentication of asset ownership when transferring an asset to a smart contract. Then, only the hash is given to the node, enabling effective anonymous asset authentication.

Reference [5] proposed a method to hide the contract address and ensure contract security. However, the candidate provided no privacy protection during the transaction, and any transaction between the two parties was publicly visible. Reference [10] defined an atomic exchange as a two-party protocol that formalizes different privacy concepts based on transaction anonymity, confidentiality, and indistinguishability. They abstract out the primitive of Atomic Release of Secrets, which captures the atomic exchange of a secret for a pre-decided transaction, uses Schnorr signatures to construct a private cross-chain swap and implements a new privacy-preserving protocol. Reference [36] presented a three-party protocol A^2L to update the payment channels in **Payment Channel Hubs (PCHs)**, using an adapter signature scheme to guarantee privacy security and improve transaction speed.

Based on the more complex notary cross-chain model, Reference [3] used the PageRank evaluation model for credit evaluation of notary nodes and designed the evaluation model to evaluate the nodes with high trustworthiness. The reliability of the cross-chain environment is ensured by dynamically eliminating possible malicious nodes through the mutual checks and balances between node evaluation and transaction evaluation.

Unlike the previous cross-chain system, Reference [38] realized the cross-chain framework using the side-chain form of main and child chains. The main chain is a licensed public chain based on relay technology, and the child chains can realize the cross-chain through the relay chain after registration. In this structure, the main chain ensures the efficient operation of the chain group and is responsible for chain group management and value anchoring. The child chains can realize the cross-chain through the relay chain after registration. Throughout the cross-chain, many solutions combine specific scenarios for privacy protection. Reference [6] proposed a cross-chain-based EHR Privacy-preserving scheme using relay chains as a cross-chain service, which ensures that patients can have secure cross-chain access to EHR data from different hospitals. Reference [12] proposed the Sunspot, a decentralized framework for data sharing that protects privacy with access control on a transparent public blockchain. It is optimized in terms of access control, and data ownership, and ensures the safety and security of shared data and sharing behavior through a standard cross-chain collaboration protocol. Sunspot solves the problem of open data key sharing, access control, and legal rights certificates on the chain in the data sharing domain.

Table 1. Agency Account‘s Expected Balance Change by Cross Chain

Agent	on Chain_a	on Relay-Chain	on Chain_b
Alice(A)	$-Pa \rightarrow Lock(+PA)$	$Note(Pa) \rightarrow Note(P*b)$	$+1 \rightarrow Lock(PB)$
Bob(B)	$+* \rightarrow Unlock(PA)$	$Note(Pa) \rightarrow Note(Pb)$	$Unlock(-PB) \rightarrow +Pb$

Among these privacy-preserving schemes, techniques such as permission control, transaction obfuscation, and zero-knowledge proof [21] are mainly used to perform cross-chain confidential transactions. In addition, there are some privacy-preserving techniques that are successfully used in blockchain and may be inspired for cross-chain, such as ring signature, mixed currency technology, address stealth [11], and other techniques for privacy information protection. In this article, we focus on combining mixed address and zero-knowledge proof techniques in a relay chain to protect the entire transaction information and property data from other cross-chain users who cannot obtain useful intelligence by tracking transactions.

3 ANONYMOUS SCHEME DESIGN

Data transfer in IoT devices can be regarded as an asset exchange between two parties. We can use blockchain cross-chain technology to explain the IoT blockchain cross-chain process, which also serves as the technical foundation for our scheme. In a cross-chain network of relay chains, we define Alice as a cross-chain user A and Bob as a cross-chain user B. The assets of one exchange unit represent the flow of assets between the two parties in the relay chain. The process is shown in Table 1, Alice initiates a transaction Pa, calls Chain_a's cross-chain smart contract to request a transfer transaction to the relay chain, and locks the Pa transaction to the public address to form a transaction PA. At this time, the relay chain locks the transactions of Alice and Bob and checks the transaction credentials from Chain_a if Alice's assets have been locked, and then it sends a cross-chain request to Chain_b, and Chain_b locks a transaction to the public account PB. The relay chain verifies the transaction credentials from Chainb again, and after both verifications are correct, it will execute the final transaction operation, transfer the unlocked assets of PB account to Bob to form the transaction Pb, complete the transaction, and write the transaction credentials $Note(Pa \rightarrow P * b)$ to the relay chain.

In this process, the increase of PA is equal to the decrease of PB, that is, $In(*PA) = Out(PB)$, but the total assets of each of Chain_a and Chain_b do not change. The relay chain completes the cross-chain transaction after two operations of collecting and verifying the transaction credentials.

The analysis of the above behavior of relay chain cross-chain transactions shows that the current problem is that, during the transaction, the relay chain network collects the credential information of both parties, and when there are other participants in the chain, the sensitive information and address information of both parties are all publicly visible in the network.

Transactions on public blockchains are typically direct address obfuscation or cryptographic channel transactions that prioritize complete anonymity and unsupervised transactions. Cross-chain transactions in pre-dominantly semi-centralized hybrid blockchains, however, cannot be fully anonymous and must be regulated and controlled.

To achieve anonymity and supervisability of transactions, we introduce a virtual address generator that encapsulates cross-chain transactions on a public address and enhances authentication mechanisms.The overall flow is shown in Figure 2. Each main transaction chain that joins this cross-chain network is connected by a relay routing router, which deploys smart contracts C for business process control in the main chain and relay chain, respectively, and smart contract C_r for transaction control in the relay chain. The following is the overall operational flow:

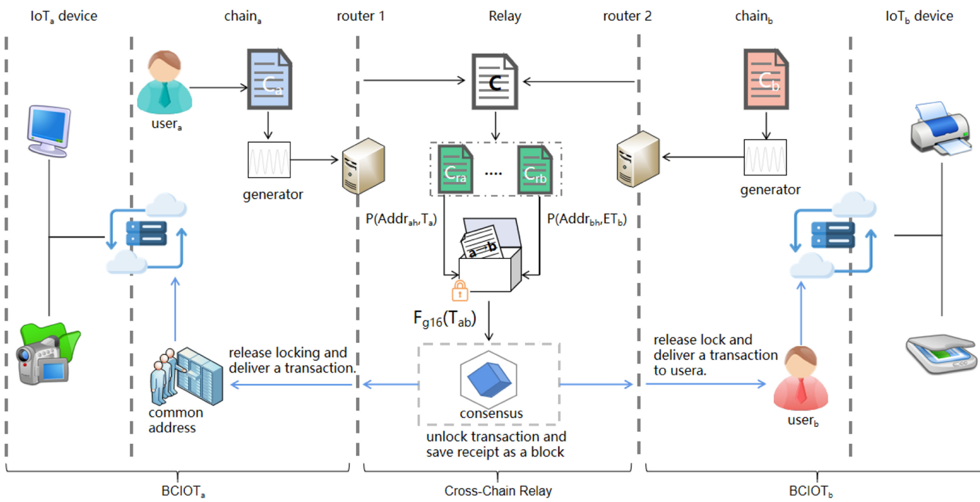

Fig. 2. Overall flow chart of relay chain privacy protection.

(1) Deploy a smart contract in $Chain_a$, which is responsible for locking assets in main chain A. When a user invokes the cross-chain contract through a client node, the value transfer to the user a is executed in $Chain_a$. It can be expressed as $-P(a) = Lock(+PA) \rightarrow Receipt(a \rightarrow A)$, which generates a transaction credential and a request to cross-chain routing Router 1.

(2) Router 1 parses the credential information and uses the Generator to generate a mapping address to the user a in Chain$_a$, creating a transaction: $P(Add_{ra}, T_a) = G_a((a \rightarrow A) \rightarrow r_a)$. This transaction has hidden the address information of user a in Chain$_a$ through the generator conversion and also under router 2 of Chain$_b$ will generate transaction $P(Add_{rb}, T_b)$, which is expressed in Relay-chain as the result of the transaction between the contract C_{ra} and C_{rb} public address.

(3) Under the control of contract C in Relay-chain, the two parties synthesize a transaction operation T_{ab} and package the transaction in the authentication module to generate credentials: $Receipt(*a \rightarrow *b) = F_{g16}(T_{ab})$, where $*a$ and $*b$ are the newly generated external addresses and F_{g16} is a representation of groth16 algorithm operations.

(4) After the relay chain has reached a consensus, relay routing unlocks both parties' transactions, records them on the chain, and completes them. Our solution is more applicable to hybrid blockchains, providing functions for supervision and transaction traceability, and its implementation is based on the generator $\varphi(x)$ function, which can implement account lookup functions and cooperative lookup functions for master-sub chain cooperation in one direction.

There are two core points in the scheme, the virtual address generator and authentication, which will be covered in Sections 3.1 and 3.2.

3.1 Virtual Address Generator

The CoinShuffle [27] scheme describes flowing between different transactions so that multiple transactions interfere with each other's paths. If a malicious node were to analyze the private data, then it would have difficulty guessing the starting and ending points of the transactions, because it cannot predict the exchange paths.

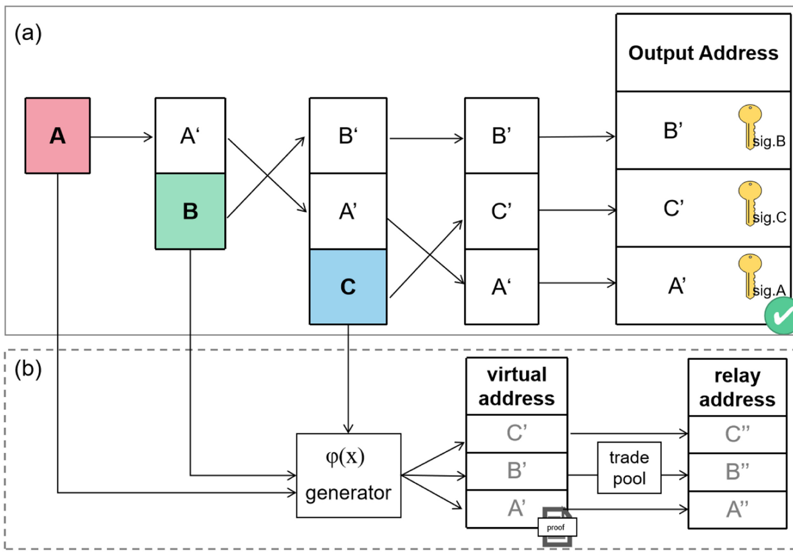

Fig. 3. CoinShuffle scheme (a) and our improved scheme (b).

The main steps in the coin mixing transaction process are depicted in Figure 3(a). First, a pool of transactions is created. Then, three transactions, A, B, and C, enter two coin mixing transactions. As a result, the address mapping between each pair of transactions is broken, making it impossible for other participants in the chain to determine which transaction should be credited to an account [35].

Our strategy, which outperforms CoinShuffle by using a virtual address generator to create external addresses directly and place transactions on a public address pool for trade, is depicted in Figure 3(b). To ensure virtual address concealment, the generating function $\varphi(x)$ is allowed to be monotonic only in a small range of the definition domain $\{x|x \in (\triangle x, \triangle x + \delta)\}$ and is an oscillating signal in a larger range. We define the simple linear relation $f(x) = \eta x + \beta$, and take a random point on this function, and the small change near its point can be expressed as $df(x) = \eta dx$, but this change is only related to the constant η, which can be evaluated by $\triangle f(x)$. The composite function $f(\sin(\omega x))$ is used based on $f(x)$ with a better result, and it is found experimentally that the composite sine function generates the point set that will make the external address exist well hidden, and we analyze its security in Section 4. The generating function $\varphi(x)$ is expressed as item (1), where the η parameter is used to control the amplitude of the function, ω controls the size of the period, β will adjust the curve in the y-axis direction, and the main role is to make the generating function in $x = 0$ when $y > 0$,

$$\varphi(x) = |\eta \cdot x \cdot \sin(\omega \cdot x) + \beta|, x \in (0, +\infty). \tag{1}$$

A discrete set of points $A(x, y)$ is taken under this signal as in Equation (2), where α is the sampling density in the x-axis and n is the number of transactions, with $n = n + 1$ for each additional transaction,

$$A(x, y) \in \left\{(x, y)|x \in \frac{1}{a} \cdot \frac{2\pi}{\omega} \cdot n, y \in |\varphi(x)|\right\}. \tag{2}$$

η, ω, and β will have a range of values that increases gradually with the number of users in the cross-chain network and is published in the cross-chain network, and each new user will take a value on a similar value domain. This process limits the curve distance between the generating functions in a small range, and attackers cannot probe user privacy via clustering analysis. Two generating functions distance $\triangle d$ is as follows in Equation (3), and the distance $d \in (\triangle d, \triangle d + n\sigma)$ between them should be made when selecting the generating functions, so as to achieve the maximum interference effect,

$$\triangle d = \sqrt{\left|\varphi_a^k(x) - \varphi_b^k(x)\right|^2 + |x_a - x_b|^2}. \tag{3}$$

Let the update operation on the coefficient vector (η, ω, β) be $\iota^i_{(\eta,\omega,\beta)}$, denoting the assignment of coefficients η, ω, β to the newly added cross-chain network i. A random coefficient vector $R_i = (\gamma_1, \gamma_2, \gamma_3)$ is selected, and the distance vector $D_i = (\triangle d_i, \triangle d_i, \triangle d_i)$ is calculated from Equation (3), resulting in the coefficient update Equation (4),

$$\iota^i_{(\eta,\omega,\beta)} = \iota^{i-1}_{(\eta,\omega,\beta)} + R_i^T + D_i^T. \tag{4}$$

In each transaction the master sub-chain takes a point in turn on the point set $A(x, y)$, which is stitched into a 64-bit external address using the snowflake algorithm, which intercepts the last 16 bits of each chain's public address as a prefix, the first bit is fixed to 0, because it is a symbolic bit, and the last 47 bits are generated by taking the first 47 bits of the y value.

Realize transaction traceability with the help of an address generator. First, query the transaction records of users on the main chain to obtain the transaction order n of the user. Then, input n into the generation function $\varphi(x)$ to obtain the point set $A(x, y)$, generate external addresses in the generator, and both sides of the transaction use the point set A_a, A_b to calculate the intersection of the following Equation (5) to obtain the list of anonymous transaction records,

$$U_{record} = A_a(x(n_a), y_a) \cap A_b(x(n_b), y_b). \tag{5}$$

Algorithm 1 shows the core algorithm of the generator, where the three parameters should be stored in the main chain and controlled by the smart contract to get the point set and the generation address.

ALGORITHM 1: Create cross-chain anonymous account

Input: A number of transaction sequence, defined as n
Output: Collection $Uvt = A(x, y)$
Set $\eta\ \omega\ \beta$ = Random values where $\omega \in (0, 1)$ and $\eta > 0$;
while *not get a generate value* **do**
    Set $x = \frac{1}{a} \cdot \frac{2\pi}{\omega} \cdot n$;
    Set y = $|\eta \cdot x \cdot \sin(\omega \cdot x) + \beta|$;
    if (x, y) *are not repeated* **then**
        Set U_{vt} = Use x,y extract m decimal places;
        return *0*;
    end
    $n = n + 1$;
end
return U_{vt};

3.2 Cross-chain Authentication

The anonymous transactions mapped by the generator may still be intercepted and tampered with as they pass through the cross-chain routing. When selecting an authentication scheme, the currently used schemes are also evaluated in this article. Reference [42] describes a cross-trust domain authentication method for digital certificate signing and distribution based on **Certificate**

Table 2. Description of the Notation

Notations	Description
$F_{g16}(P, V, W)$	Groth16 algorithm generating function
$Proof_i$	Proof of Groth16 algorithm
GPk_i, GVk_i	The prove key and verify key of i
Pk_i, Sk_i	The public key and private key of i
$Ep(a, b)$	An elliptic curve over finite field
ID, χ_m	Blockchain id and a message under m purposes
$\|\|$	Data concatenation
W, S	Algebraic closure in the domain F^{m-l} and domain F^l
$Addr_{mi}$	The i blockchain address under m purposes
$Split(ST, N), Swap(a, b)$	Split string ST by length N and Swap a, b
TX^i	A blockchain transaction
$Enc(Pk_i, M), Dec(Sk_i, M_1)$	Encryption of plaintext M, and Decrypt the message M_1

Authentication (CA) nodes. Reference [7] using a blockchain as a certificate distribution storage authority, taking advantage of the blockchain's chained storage and tamper-evident nature, in which users extract certificates and public keys directly from the blockchain to verify their identity. Reference [41] incorporates registration authentication into the blockchain consensus to store the signed information in the blockchain.

In cross-chain transactions, the attacker identifies a user in the network by a certificate hash. In this article, we use zero-knowledge proofs based on the framework in Reference [41], improving the groth16 algorithm verification logic so that it can generate an additional determining common parameter to verify the polynomial held by the prover. In this non-interactive verification process, neither the prover nor other third parties can guess and falsify the answer, which will be able to better accommodate the authentication process across the chain network, without requiring any additional resources. The following three steps summarize the core of the theory of the groth16 [16] algorithm:

(1) Randomly selected the parameters: $\alpha\ \beta\ \gamma\ \delta\ , x \leftarrow F*$, statements is $(a_1, \ldots, a_\ell) \in F^\ell$, witness is $(a_{l+1}, \ldots, a_m) \in F_{m-\ell}$, calculate $\tau = (\alpha, \beta, \gamma, \delta, x)$,

$$\begin{cases} \psi_1 = \left\{x^i\right\}_{i=1}^{n-1} \\ \psi_2 = \left\{\frac{\beta u_i(x)+\alpha v_i(x)+w_i(x)}{\gamma}\right\}_{i=0}^{l} \\ \psi_3 = \left\{\frac{\beta u_i(x)+\alpha v_i(x)+w_i(x)}{\delta}\right\}_{i=l+1}^{m} \\ \psi_4 = \frac{x^i t(x)}{\delta}{}_{i=0}^{n-2} \\ \sigma = (\alpha, \beta, \gamma, \delta, \psi_1, \psi_2, \psi_3, \psi_4) \end{cases}, \tag{6}$$

(2) Select two parameters: r and s, and calculate $proof\,\pi = \prod \sigma = (A, B, C)$, which:

$$\begin{cases} A = \alpha + \sum_{i=l+1}^{m} a_i u_i(x) + \gamma\delta \\ B = \beta + \sum_{i=l+1}^{m} a_i u_i(x) + s\delta \\ C = \frac{\sum_{i=l+1}^{m} a_i(\beta u_i(x)+\alpha v_i(x) w_i(x))+h(x)t(x)}{\delta} \end{cases}. \tag{7}$$

(3) Calculate whether the equation holds whether the prover is correct, as in the following equation:

$$A \cdot B = \alpha \cdot \beta + \frac{\sum_{i=l+1}^{m} a_i(\beta u_i(x) + \alpha v_i(x) w_i(x))}{\delta} + C \cdot \delta. \tag{8}$$

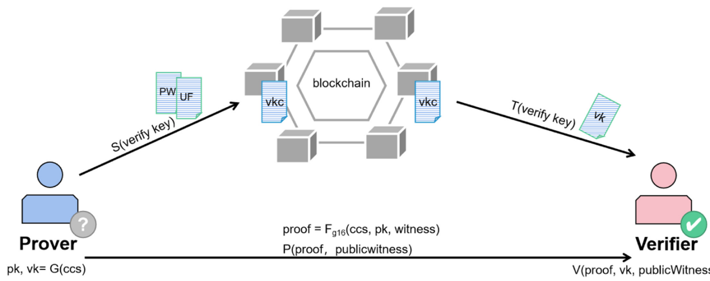

Fig. 4. Authentication scheme using zero-knowledge proof.

Based on this algorithm, we have made some modifications to make the authentication process more suitable for cross-chain services; the authentication process is depicted in Figure 4, and it is divided into five major steps:

(1) The proof polynomial is transformed into a constraint circuit, in which the polynomial is reduced and orderly transformed into a set of three vectors A, B, and C, and the constraint satisfies $A \cdot s \times B \cdot s = C \cdot s$, where s corresponds to the vector computed by each arithmetic gate circuit.
(2) Calculate three finite groups G_1, G_2, G_T, corresponding generating elements are g, h, $e(g, h), GPk = g^y, GVk = h^y$, where Vk is generated and will be stored to the relay chain.
(3) Extract the common part of the witness and generate the evidence: $Proof = F_{g16}(G_1, G_2, G_T, Pk, witness)$.
(4) When the Prover proves his identity to the Verifier, evidence $Proof$ is generated and this information will be sent to the Verifier.
(5) The verifier obtains $Proof$ and gets the corresponding GVk of the prover by address query and then verifies the identity of the prover by using the bilinear pairing function if it is correct.

Our scheme uses the generating function to create a constraint system, which is then transformed into the QAP problem of the groth16 algorithm. The calculated point set $A(x, y)$ is entered into the system to obtain evidence about the virtual address, which is saved on the relay chain along with the entire cross-chain transaction process, providing non-interactive proof for transaction authentication and traceability while also ensuring data security in the communication process.

3.3 Cross-chain Privacy Protection Protocol Design

In this section, an interaction protocol for cross-chain privacy protection is proposed. Initialization and transaction are the two phases that make up the protocol. The initialization step primarily produces authentication files, which are utilized in the encrypted transmission and identity verification processes and stored in the relay chain network. Asset transfer and transaction content determination are carried out during the cross-chain transaction stage. The entire protocol is as follows:

(1) Initialization stage: Choose a large number k to generate the private key Sk, choose a base point p on the elliptic curve $Ep(a, b)$ to generate the public key $Pk = Sk \cdot p$, and generate the proof keys GPk, GVk according to the steps of the groth16 algorithm described in Section 3.2.
(2) Transaction stage: In this stage, the cross-chain resource exchange is mainly carried out, and its interaction steps are expressed as shown in Figure 5.

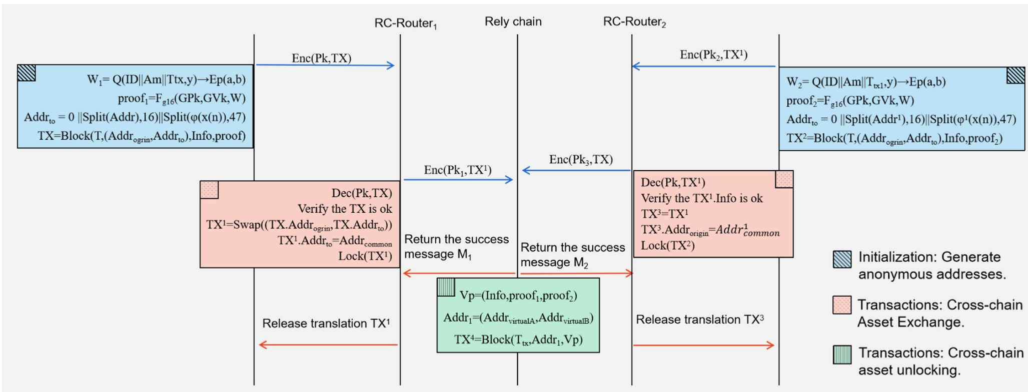

Fig. 5. Cross-chain interaction protocol.

(a) The transaction initiator selects a polynomial $Q(x, y)$. Using the $Q(ID \parallel Am \parallel T_{tx}, y)$ to find the polynomial value y, and the proof is computed and generated according to the algorithm described in Section 3.1,

$$\begin{aligned} W_1 &= Q(ID \parallel \chi_m \parallel T_{tx}, y) \rightarrow Ep(a, b) \\ proof_1 &= F_{g16}(GPk, GVk, W). \end{aligned} \tag{9}$$

(b) The generator algorithm in Section 3.1 of this section is used to select a generator function $\varphi(x(n))$ to generate a virtual address. The initial content of the transaction data block includes the current time, local address, destination address, and transaction information. The destination address is the mapped virtual address, and the complete transaction data block is calculated as follows:

$$\begin{aligned} Addr_{to} &= 0 \parallel Split(Addr_{common}), 16) \parallel Split(\varphi(x(n)), 47) \\ TX &= Block(T, (Addr_{ogrin}, Addr_{to}), Info, proof). \end{aligned} \tag{10}$$

(c) Send the transaction TX to the cross-chain router for verification, and then swap the local address and destination address. Finally, replace the destination address with the common address to create a new transaction,

$$TX^1 = Swap((TX.Addr_{ogrin}, TX.Addr_{to})). \tag{11}$$

Then send $TX^* = Enc(Pk, TX_1)$ to the relay chain. In this process, the receiver performs the same steps as the initiator. There are some differences in that the receiver accepts the transaction and executes the transfer operation from the public address to the user address.

(d) The relay chain requests the release of resources from the receiver after decrypting and verifying the transaction content TX^4, exchanging the local address, and replacing the destination address with the receiver's address.

4 PRIVACY AND SECURITY

In a zero-trust BCIOT environment, we cannot accurately determine the loyalty of each node. Malicious devices accessing the network will cause data tampering, information loss, and privacy leakage. Ensuring privacy and security in an autonomous IoT environment are consistent with the blockchain requirements for the transaction environment [8]. Data security requires cross-chain solutions to guarantee proper cross-chain transactions in terms of availability, confidentiality, and integrity [17]. Availability needs to meet the user's demand for data usage and still be recoverable in the event of external attacks on the system. Integrity requires that transaction data not be tampered with during transmission and uploading. Confidentiality ensures that each transaction is controlled by authority and that the data is not globally visible.

4.1 Protocol Security Analysis

(1) Anti-replay attack: Our protocol uses $proof$ to verify the identity of the owner of a transaction mapped to a virtual address. An attacker must first read the evidence field W to intercept proof, whereas the original transaction W makes use of the timestamp T_{tx} created at the moment of the transaction. The time of the transaction and the computed proof time are not equal at the time of verification, even if the attacker assumes the identity. The fake identity will eventually be unable to be verified.

(2) Anti-Man-in-the-Middle Attack: We can assume that the attacker counterfeits the relay chain and copies partial information of the relay chain. When sending a transaction to the forged chain, both parties use the public key generated during the initial registration with the federated chain to establish an encrypted channel. However, the attacker does not know the corresponding private key, the verification does not pass, and the Transaction will not succeed. Suppose the attacker directly forges the transaction party or receiver. Before sending transaction information to the relay chain, the transaction initiator and receiver generate their transaction blocks TX_a, and TX_b according to the rules in Equation (10). Yet, the $proof$ in the transaction block generated by the groth16 algorithm is unique and can be verified using the GVk, which is stored on the relay chain. After both sides of the transaction verify each other's transactions, the transaction fails.

(3) Security of zero-knowledge proofs: The core of zero-knowledge proofs is the problem of polynomial factorization. Find a polynomial $H(x)$, satisfying $C(x) \cdot s - A(x) \cdot B(x) \cdot s = H(x) \cdot Z(x)$ in an algebraically closed domain, and the prover computes $P(x)$ with the held solution s. $H(x) = P(x)/Z(x)$, and the verifier verifies whether $P(x) = H(x) - Z(x)$ holds and is able to know whether the prover holds the answer. To make it impossible for the provers to falsify $P(x)$ and $H(x)$, the point p is generated by random sampling in a function $\varphi(x(n))$, using homomorphic encryption to process the set of points as a mapping of $E(\xi^i)(E(p^1), E(p^2), E(p^3)......E(p^n))$, and finally, the verifier checks that $E(P(p)) = E(H(p) - Z(p))$ verifies the legitimacy and the proof is secure as the provers do not know the mapping relation and cannot falsify the evidence.

(4) Anonymity: We define two users a, and b, where a is the initiator, located in blockchain A, and the recipient b is located in blockchain B. The transactions are performed on path Γ according to our ross-chain interaction protocol. Representing the anonymous interaction process as a graph path with vertices v_i, $i = 1, 2, 3, 4, 5$ and edges e_j, $j = 1, 2, 3, 4$, the set of edges $\Gamma(a, b)$ from a to b is given as follows:

$$\Gamma = \begin{cases} e_1 = < v_1, v_2 > = < Addr_a, Addr^a_{virtual} > \\ e_2 = < v_2, v_3 > = < Addr^a_{virtual}, Addr_{common} > \\ e_3 = < v_3, v_4 > = < Addr_{common}, Addr^b_{virtual} > \\ e_4 = < v_4, v_5 > = < Addr^b_{virtual}, Addr_b > \end{cases}. \tag{12}$$

The paths Γ_r, Γ^+, Γ^- can be resolved in the relay chain and blockchain A, B respectively, and the protocol in this article protects the information flow between the two sides of the transaction by splitting the two transactions into three segments. The relay chain only participates in one of the transactions and cannot obtain the complete information of

Table 3. Comparison of the Protection Effectiveness of Different Schemes against Several Privacy Attacks

Attack Type	LSTM-TC [35]	PCHs [36]	CoinShuffle [27]	Ours
Dust Attack	×	×	√	√
Clustering Analysis	√	√	×	√
Centralization issue	√	×	—	—
Flow Analysis	√	—	√	√
Fingerprint Value	×	×	×	√
Round-trip trading Attack	×	×	√	√

"√" means that the attack is currently preventable, "×" means that there is currently no countermeasure, and "—" means that the problem does not exist in this scheme.

the user,

$$\Gamma_r = e_2 \wedge e_3, \Gamma^+ = e_1 \wedge e_2, \Gamma^- = e_3 \wedge e_4. \tag{13}$$

The mapping relationship between v_1 and v_4 depends on the generating function $\varphi(x)$ and the confidential identity ID. To access the user information, the attacker must first identify the chain where the user is situated, enter the internal blockchain A to read the block, find the association information, and then exhaustively enumerate according to the rules of the generating function. Finally, they can find a key path (Γ^+, Γ_r) or (Γ_r, Γ^-). The attacker will not, however, be able to enter due to the identity verification access constraints of the federated chain. An attacker cannot access the crucial transaction path, $\Gamma_{keypath} = \{\emptyset\}$, which ensures the confidentiality of cross-chain transactions and cannot hack into the adversary's network.

4.2 Privacy Protection

We focus on address mapping, improve the cross-chain transaction process based on CoinShuffle, and split the correlation between transactions. Unauthorized nodes cannot effectively infer whether two transactions have backward and forward continuity or whether they belong to the same account. The coin-mixing protocol is most vulnerable to malicious infiltration attacks by certain participating members. On the one hand, our scheme is designed based on the hybrid blockchain, in which the transaction authorization and access control are stricter and can filter most of the malicious nodes. On the other hand, transactions sent to the relay chain were replaced with common addresses, and the generation function will provide virtual addresses for mapping, corresponding to multiple non-repeating anonymous external addresses for each transaction. To infiltrate the attack maliciously, a malicious user must first breach the main chain and obtain all transaction block records to infer the user's transaction privacy information.

Table 3 compares the effectiveness of several privacy protection schemes against privacy attacks. All three comparison schemes have vulnerabilities in dealing with value fingerprint-based attacks, and the impact of value fingerprinting can only be eliminated in conjunction with other technical means. The solution in this article is to slice the transaction twice to form multiple scopes, and the value fingerprint cannot be continuously passed on through the transaction.

As cross-chain system communication becomes more complex, there is the problem of man-in-the-middle attacks. Incalculable losses will be incurred if a malicious user hijacks the cross-chain route and modifies the transaction information. Our scheme utilizes encrypted communication and also changes part of the authentication process, relying on the security of the groth16 algorithm, so that no modification can be made even if the relevant information is intercepted. Because a transaction on the relay chain requires both parties to provide generated evidence to prove the

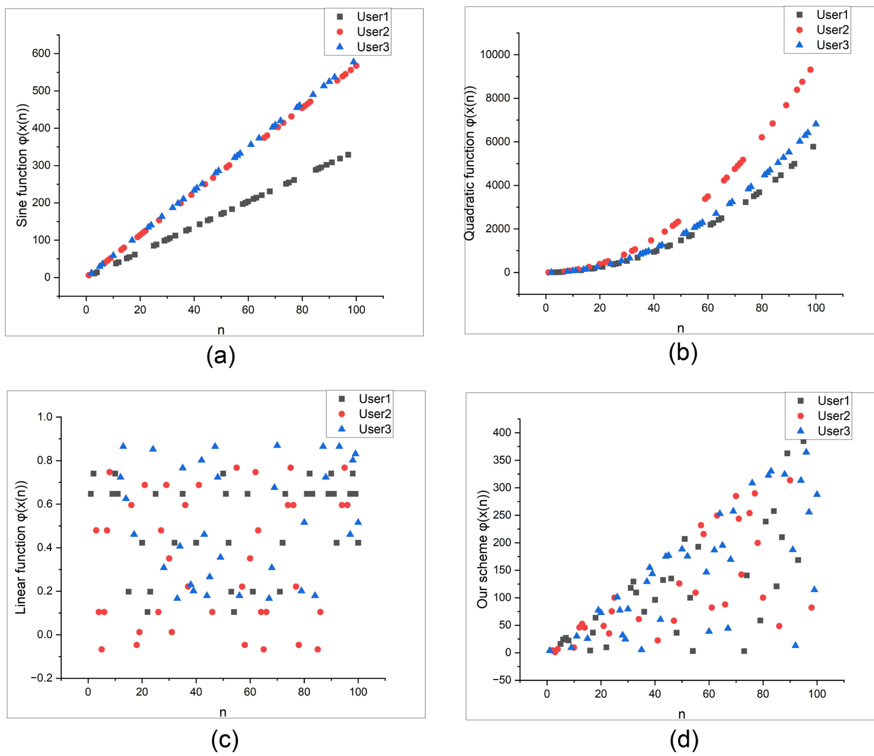

Fig. 6. Relationship between anonymous address and transaction order n in different generating functions.

attribution of the transaction, the verification keywords are stored on the chain and cannot be changed.

In our scheme, the generating function $\varphi(x(n))$ associates user transactions with address information using discrete points, and each user maps to an infinite number of addresses. When various generating functions are applied, a plot of the link between the generated values and the order of user participation in the transaction n is shown in Figure 6. The created points in Figure 6(c) are dispersed more equally in space, since only one sine function is used as the generating function, but when more samples are taken, more duplicate address values are produced, which slows down transactions. In Figure 6(a) and (b), simple linear and quadratic functions are used, corresponding to a clear correlation between the generated addresses and the order of the transaction. Figure 6(d) makes use of a nonlinear composite function to limit the distribution area of spatial points, and reasonable parameters can be used to prevent duplicate addresses. The generating function utilized in this article's system ensures that user transaction linkages cannot be analyzed, because the relationship between user participation transactions and produced addresses are no longer linear and the genuine correspondence is random.

On the cross-chain system, multiple sets of generating functions exist for each blockchain, which is stored on the local cross-chain nodes to provide external computation. The entire privacy protection system remains secure as long as this sequence of generating functions and the main chain record are not violated. The generating function is controlled by four parameters, and the original point set is neither used directly during the transaction nor can the mapping relationship of the transactions on the chain be analyzed by regression or clustering, ensuring that user privacy information is not leaked.

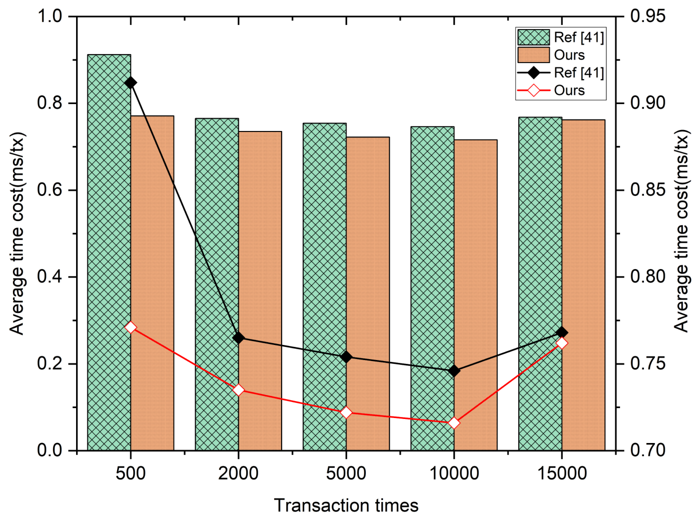

Fig. 7. Average cost time of different identity authentication schemes in 500, 2000, 5000, and 10,000 transaction times.

5 PERFORMANCE

In this article, we design three experiments from storage overhead, transaction speed, and compare the performance and stability with the original scheme, as well as analyze its security. The experimental system environment is Linux 4 core 4G, three Fisco Bcos federated chains are used to build a cross-chain environment, and the cross-chain routing function is simulated using go-SDK access.

The authentication link was first tested, and we ran tests of the CA scheme based on Reference [42] and our improved scheme in the same environment, which was mainly used to verify the performance advantages of the scheme in this article in a relay chain cross-chain environment.

Figure 7 shows the average cost time of different identity authentication schemes in 500, 2,000, 5,000, and 10,000 transactions run in the same experimental environment. We can see a 3% improvement in time between the base scheme and our improved scheme, and the verification stability is also improved compared to the original scheme. The running time difference between the two schemes shown in the figure is small, but after counting the standard deviation, we can find that the scheme [42] fluctuates a lot, mainly because of the frequent interaction process and the long time occupied by the public and private key transmission, and the change to non-interactive proofs to reduce the communication process; second, the I/O operations and encryption/decryption operations are reduced, which can optimize some time costs.

Figure 8 shows the storage and transfer size statistics for the whole verification process. It can be seen that the improved scheme has greater progress in local storage, on-chain storage and transmission bytes, and the smaller transmission size will have a greater advantage in long-range high-latency networks.

Figure 9 shows the comparison of transaction time under the different numbers of users. We can see that the coin-mixing scheme can provide better performance in the initial situation when the number of users is less than 15, but as the number of users in cross-chain transactions increases, the transaction time increases sharply due to the complexity of mixing coin technology in operation. The performance of our scheme is relatively smooth, and when the number of users reaches 50, its single transaction time is only 92.4 ms, and the difference between the minimum and the maximum

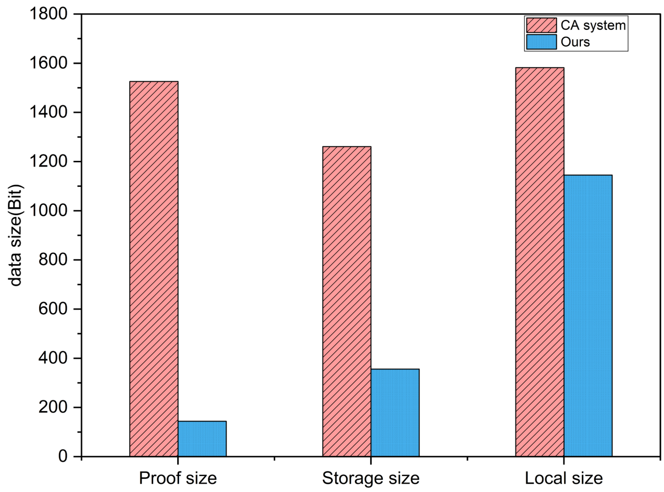

Fig. 8. Comparison of the storage size and transfer size in the validation process.

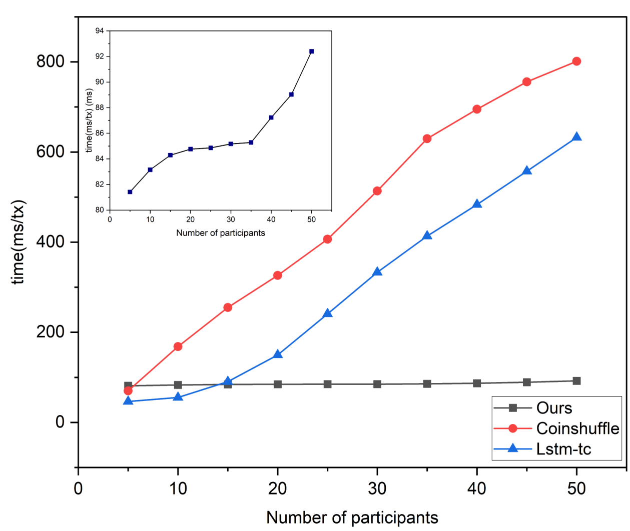

Fig. 9. Change in transaction time as the number of users changes under different schemes (the small graph shows the trend of time change in the increase of users in the network under our scheme).

number of users in the experiment is less than 10 ms, which is much smaller than that of the coin mixing scheme in References [13, 27].

6 CONCLUSION

The massive amount of data generated in the IoT has developed into a significant commercial asset. Controlling access rights is quite challenging when sharing data in zero-trust IoT blockchains via cross-chain due to the involvement of multiple nodes. Insecure network environments can result in the release of sensitive data.

To solve the issue of cross-chain privacy leakage in a zero-trust IoT environment, we lead into the groth16 generating function and zero-knowledge proof algorithm and propose a generic relay chain cross-chain privacy protection scheme. It achieves cross-chain anonymous transactions and supervisable functions on a hybrid blockchain. Furthermore, we make secure and controllable adaptive modifications to the generating function and identity authentication. Experimental shows that its transaction time is shorter than that of the address obfuscation technique, and the storage size and transmission size of transaction proofs are smaller. And the functionality can be achieved using multiple smart contracts, which are compatible with existing framework standards and can effectively guarantee the cross-chain privacy of the IoT blockchain.

In this article, we achieve the hiding of key user information but given the amount of information generated by both parties' asset anchoring before the transaction still has some privacy risk. The future work will focus on changing the relay chain consensus method, using homomorphic encryption to operate after encrypting the asset information, and optimizing the framework of this article to speed up the transaction time and reduce the number of transaction communications.

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